Q1. What is the value of result?

python
```
result = 2 ** 3 ** 2 + 10 // 3 * 2 - 4 % 3
```
A) 512
B) 517
C) 519
D) 522

Q2. Evaluate this bitwise-logical hybrid:

python
```
x = 5
y = 2
z = 3
print(x & y | z ^ x << 1)
```
A) 9
B) 13
C) 15
D) 7

Q3. What does the following print?

python
```
print(not 5 > 3 & 2 or 4 ** 2 // 3 + ~1 & 7)
```
A) False
B) True
C) 11
D) 12

Q4. What is printed?

python
```
a = 256
b = 257
print((a is b - 1) == (a == b - 1))
```
A) True
B) False
C) TypeError
D) SyntaxError

Q5. What is the final value of res?

python
res = ~(5 + 3) << 2 & 7 or not 10 // 3
A) 4
B) 0
C) True
D) 1

Q6. What is the type and value of result?

python
result = 0 or "Python" and [] or 42
A) 42 (int)
B) [] (list)
C) "Python" (str)
D) True (bool)

Q7. How many times is lst.pop() executed, and what is the final value of result?

python
lst = [10, 20, 30, 40]
x = 0
result = (x > 0) and (lst.pop() > 15) or (lst.pop() < 25)
A) 1 call, True
B) 2 calls, True
C) 1 call, False
D) 2 calls, False

Q8. What is printed?

python
print(5 or 0 and 2, 0 and 5 or 2)
A) 5 2
B) True True
C) 5 0
D) 0 2

Q9. Why does this loop never terminate, and what is the fix?

python

```
i = 0
while i < 10:
    if i % 3 == 0:
        continue
    print(i)
    i += 1
```

A) i is never incremented when i%3==0; fix by moving i += 1 before the if.
B) The condition i < 10 is always true; fix by changing to <=.
C) continue is invalid inside while.
D) print(i) raises an error.

Q10. Which of the following expressions does NOT raise a ZeroDivisionError?
A) 10 // 0 or 5
B) 0 or 10 // 0
C) 0 & 10 // 0
D) 10 // 0 and 5

Q11. What is the output of:

python

```
print(-17 // 4, -17 % 4, 17 // -4, 17 % -4)
```

A) -5 3 -5 -3
B) -4 -1 -4 1
C) -5 3 -4 1
D) -4 3 -5 -3

Q12. Compute ~-5 and ~5 respectively:
A) 4, -6
B) -4, 6
C) 4, 6
D) -4, -6

Q13. What is the integer output of:

python

```
print((~0b1010) & 0b1111, (-5 >> 1) & 0b111)
```

A) 5, 3
B) 5, 5
C) 10, 3
D) 10, 5

Q14. Verify the Euclidean identity: a = b * (a // b) + (a % b).
Which of the following is FALSE for a = -10, b = 3?

A) a // b is -4
B) a % b is 2
C) 3 * -4 + 2 = -10
D) a // b is -3

Q15. What is printed?

python
print(True + True, False * 3, ~False & True)

A) 2 0 1
B) 2 0 0
C) 1 0 1
D) 2 3 0

Q16. What is printed?

python
lst = [1, 2]
print(1 in lst == True)

A) True
B) False
C) TypeError
D) ValueError

Q17. How many times is pop() called in this chain?

python
lst = [1, 2, 3, 4]
print(lst.pop() < lst.pop() < lst.pop())

A) 1
B) 2
C) 3
D) 4

Q18. (Memory interning) What is printed?

python
a = 1000
b = 1000
c = a
d = a + 0
print(a is b, a is c, a is d)
A) False True False
B) False True True
C) True True True
D) False False False

Q19. What is printed?

python
s = "abcabc"
print("ab" in s, "ab" in s[1:], s[1:] is s)
A) True True False
B) True False False
C) False True False
D) True True True

Q20. (Chained comparison with mutation) What is printed and what is the final lst?

python
lst = [10, 5, 0]
print(1 < lst.pop() < 6)
print(lst)
A) True [10, 0]
B) False [10, 0]
C) True [10, 5]
D) False [10, 5]

Q21. What is the output of this loop?

```
python
for i in range(3):
    if i == 1:
        continue
    print(i, end=" ")
else:
    print("Done")
```

A) 0 2 Done
B) 0 1 2 Done
C) 0 2
D) 0 1 2

Q22. What is the total value of count after execution?

```
python
count = 0
i = 0
while i < 3:
    j = 0
    while j < 3:
        count += 1
        if j == 1:
            break
        j += 1
    i += 1
print(count)
```

A) 3
B) 6
C) 9
D) Infinite loop

Q23.
Assertion (A):
a = []
b = a
x = a or b and [1]
a.append(1)
print(x is a)
print(x == a)

Reason (R):
and has higher precedence than or, and a or (b and [1]) evaluates to the first truthy operand without necessarily evaluating every operand.

A. Both A and R are true, and R correctly explains A
B. Both A and R are true, but R does not correctly explain A
C. A is true, but R is false
D. A is false, but R is true

Q24.
Assertion (A):
x = 5
result = 1 < x < 10 == x
is equivalent to:
result = (1 < x) < (10 == x)

Reason (R):
Python's chained comparison:
a < b < c
is evaluated as:
(a < b) and (b < c)
and b is evaluated only once.

A. Both A and R are true, and R correctly explains A
B. Both A and R are true, but R does not correctly explain A
C. A is true, but R is false
D. A is false, but R is true

Q25.
Assertion (A):
x = 6
y = 3
print(x & y == 2)
prints:
True

Reason (R):
Bitwise & has higher precedence than comparison ==, so Python evaluates:
(x & y) == 2

A. Both A and R are true, and R correctly explains A
B. Both A and R are true, but R does not correctly explain A
C. A is true, but R is false
D. A is false, but R is true